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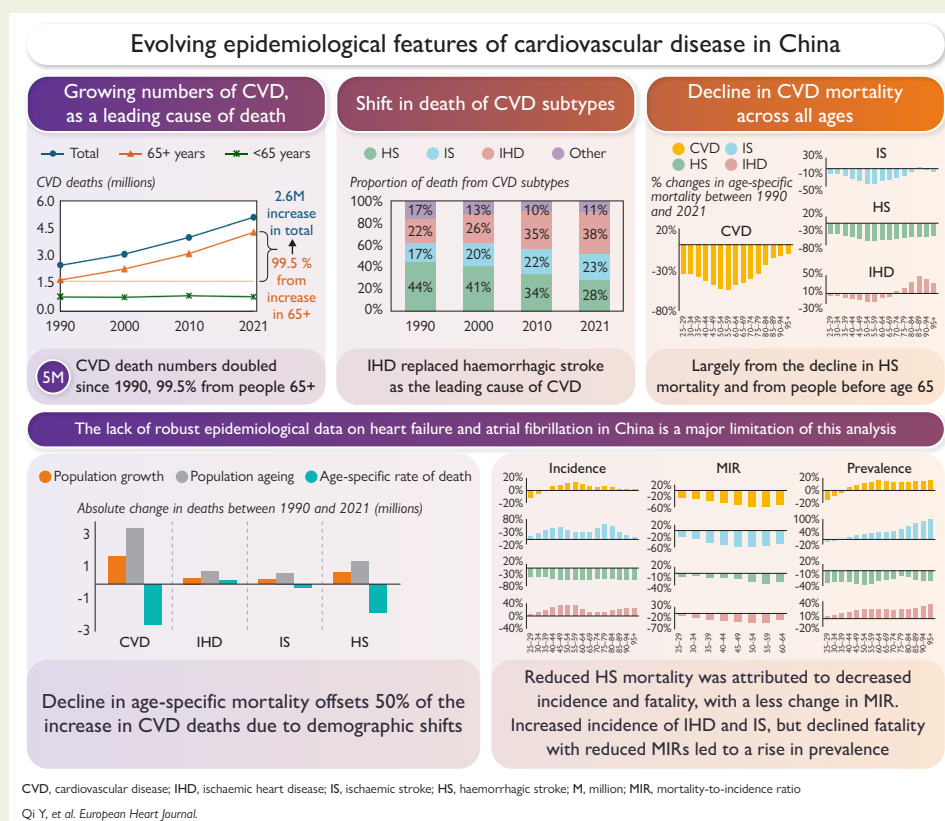
Cardiovascular disease in China: epidemiological evolution and implications

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Graphical Abstract



Evolving epidemiological features of cardiovascular disease in China. A visual summary of the evolving features of the CVD epidemic in China over three decades. CVD includes IHD, IS, HS, and other types. HS includes intracerebral haemorrhage and subarachnoid haemorrhage. Data for this analysis were sourced from the Global Burden of Disease study (<http://ghdx.healthdata.org>). Mortality-to-incidence ratio, defined as an alternative indicator of the case fatality rate, was calculated as the proportion of age-specific mortality relative to the corresponding incidence rates of a particular disease over a 1-year period. The percentage change of the 2021 rate relative to the 1990 rate is calculated as $(2021 \text{ rate} - 1990 \text{ rate}) / 1990 \text{ rate} \times 100\%$. The method of decomposition analysis is based on the methodology described by Roth *et al.* Detailed methods were in the [Supplementary Materials](#).

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Abstract

Cardiovascular disease (CVD) has remained a predominant cause of mortality in China for several decades, experiencing notable epidemiological transitions. Numerous recent studies have provided valuable observational data on the evolution of CVD epidemiology across various dimensions. However, there is a paucity of comprehensive reviews that synthesize these findings and analyse their interrelationships. To address this gap, this review summarizes the key features from complex data presented in various literature reports and additional analyses of available databases. The impact of the primary drivers of these changed features and their intricate interconnections on total CVD and major subtypes, including ischaemic heart disease (IHD), ischaemic stroke (IS), and haemorrhagic stroke (HS), was quantitatively evaluated across different periods from 1990 to 2021. Four prominent transitional features of CVD mortality and the impact of underlying drivers elucidate not only new challenges for CVD prevention and control but also highlight the hidden effects of national efforts on CVD prevention. The analysis also indicated that despite a similar pattern in age-specific mortality for HS, IS, and IHD over the past decade, only the decline in HS mortality was largely attributed to the benefits of primary CVD prevention. In contrast, the declining age-specific IHD and IS mortality was due to reduced fatal events from improved medical care. The implications of these evolving features of CVD over time for further decision-making in CVD prevention strategies are discussed in depth.

Keywords Cardiovascular disease • Evolving epidemiology • Ageing • Metabolic risk factors • Quality of care

Introduction

In China, cardiovascular disease (CVD) has remained a major threat to population health as a leading cause of death and disability.^{1–3} However, the landscape of CVD has undergone a dynamic transformation, driven by the rapid socioeconomic development and the associated changes in demographics, lifestyle patterns, healthcare infrastructure, and national efforts on CVD prevention and treatment over the past decades. Understanding the evolving epidemiological features, along with insights into the impact of positive or negative driving factors, is critical for accurately evaluating the achievements and limitations in CVD prevention and treatment, informing decision-making, and ultimately improving cardiovascular health at the population level.

In recent years, numerous publications offered valuable observational evidence on the evolution of CVD epidemiology in multifaceted aspects, including temporal and spatial changes in the CVD burden,^{4–8} updated profiles of risk factors,^{3,9–27} and advances in national policies of CVD prevention and treatment.^{28–31} Although several comprehensive reviews were published 6 years ago to summarize the features of CVD epidemiology in China, the rapidly evolving changes and their implications for decision-making in CVD prevention require updating.^{6,32} Furthermore, few comprehensive reviews have summarized the quantitative impact of the underlying drivers on the changes in mortality of CVD and subtypes in the whole population, as well as among different age groups and by sex. To address this gap, this review aims to summarize not only the features of epidemiological transition but also to evaluate the impacts quantitatively from the underlying drivers and their intricate, interconnected effects and discuss the implications with insightful perspectives.

To provide convincing information on this topic, we prioritized the data extracted from high-quality databases when available, along with the information from published papers. Therefore, the references and data presented in this review are primarily derived from three sources. The first is the publications indexed on PubMed and relevant books (both in English and Chinese) that focus on recent epidemiological studies related to CVD, risk factors, or studies on CVD prevention and treatment in China. The second source was the open database

of the Global Burden of Disease (GBD) study.³³ The third data source was the NCD Risk Factor Collaboration (NCD-RisC) database.³⁴

Evolving epidemiological features of cardiovascular diseases in China

This review summarized four significant evolving features of CVD deaths across different periods from 1990 to 2021 in China. The Graphical Abstract provides a visual summary.

Feature one: the high CVD absolute death burden is predominantly from ageing people

In 2021, CVD caused 5.1 million deaths in China, accounting for more than one-fourth of global CVD deaths and nearly half of total deaths in China. From 1990 to 2021, CVD deaths increased by 2.6 million, with 99.5% of this rise occurring in population aged ≥ 65 —the elderly people as defined by the World Health Organization. The proportion of CVD deaths aged ≥ 65 in total CVD deaths increased from 63.7% to 80.6% in men and from 72.3% to 88.4% in women between 1990 and 2021 (*Graphical Abstract*).

Feature two: further increasing burden from atherosclerotic CVD, including ischaemic heart disease and ischaemic stroke

Of all CVD deaths across all ages in the year of 2021, 61% were attributed to atherosclerotic CVD. The increase in atherosclerotic CVD occurred more rapidly from 2000 to 2010, during which time China transitioned from a low-middle-income country to a high-middle-income country. Ischaemic heart disease (IHD) became the top cause of CVD death in 2010 and became the predominant cause of CVD death in 2021. The features of changes were in a similar pattern and extent in men and women. Additionally, the proportion of deaths from other types of CVD declined over time (*Graphical Abstract* and *Supplementary data online, Figure S1A*).

Feature three: declining age-standardized mortality and increasing crude CVD mortality

Between 1990 and 2021, the age-standardized CVD mortality rates decreased by 31%, with an estimated annual percentage change (EAPC) of -1.14% (95% confidence interval: -1.32% , -0.96%) (Methods in [Supplementary Material](#)). The most striking decline was from the age-standardized mortality of haemorrhagic stroke. The age-standardized rate of CVD deaths before age 65 fell by 37.5% (from 142.8 to 87.2 per 100 000 population) in men and by 64.1% (from 123.6 to 43.7 per 100 000 population) in women. This decline reflected the combined effects of improved risk factors management in primary prevention, higher survival after acute CVD events, and a decreased risk of recurrent CVD. However, the crude CVD mortality rose by 68% (EAPC: 1.77% , 95% confidence interval: 1.66% , 1.88%) in the same period, mainly due to the increased deaths from IHD and ischaemic stroke in the population aged ≥ 65 (see [Supplementary data online, Figure S1B and C](#)).

Feature four: a comprehensive decline in age-specific CVD mortality across all age groups from age 25 to ≥ 95 years

There was a substantial decline in age-specific CVD mortality across all age groups (25 to ≥ 95 years) except for men aged over 95 years between 1990 and 2021 ([Graphical Abstract](#) and [Supplementary data online, Figure S1C](#)). This decline was driven mainly by the significant reduction of haemorrhagic stroke mortality. Women had a more profound decline in total CVD and haemorrhagic and ischaemic stroke mortality in all age groups than men but a bidirectional change in IHD mortality, with a decrease in those aged < 75 years and an increase in those aged ≥ 75 years. In men, age-specific IHD mortality increased in most age groups, especially among the elderly. Despite the declining trend in age-specific CVD mortality, the age-specific proportion of CVD deaths in all-cause deaths remained in similar patterns, with a slight increase in proportion among population aged 70 and above (see [Supplementary data online, Figure S2](#)).

The evolving features of CVD mortality in China were shaped by the synthetic effects of underlying determinants of three aspects, including demographic shifts, changes in CVD incidence, and case fatality. Particularly, age-standardized or age-specific CVD incidence and case fatality rates reflect the performance of primary prevention, secondary prevention, and emergency care capacity in China, influencing both CVD mortality and prevalence.^{35–39} Assessing the different impacts of these determinants provides a valuable reference for decision-making regarding CVD prevention strategies and resource allocation in China.

Evaluation of impacts from underlying drivers for the evolving epidemiological features of CVD

The impact of demographic shift

China's demographic landscape has undergone dramatic changes over the past three decades. The proportion of its ageing population increased from less than 6% in 1990 to 14% in 2021 (see

[Supplementary data online, Figure S3](#)), with projections reaching 26% by 2050.⁴⁰ This ageing trend has significantly influenced the evolving CVD burden in China,^{38,39,41,42} contributing to a 64% increase in CVD-related deaths between 2005 and 2020 as reported in a recent study.³⁹

Demographic shifts, characterized by the increased proportion and absolute numbers of both aged population and all age population, resulted in an additional 5.2 million CVD deaths between 1990 and 2021, with 3.5 million (67%) attributed to a change in age structure, and 1.7 million (33%) to population growth ([Figure 1](#), methods in [Supplementary Material](#)). Decrease in age-specific CVD mortality, driven by the combined efforts in CVD prevention and treatment, led to 2.6 million fewer CVD deaths, thereby offsetting 50% of the increase attributable to demographic transitions. Moreover, 37% (0.82 million) of the decreased CVD deaths occurred in individuals aged under 65. However, the reduction in CVD deaths attributable to CVD prevention and treatment strategies was completely counterbalanced by rising CVD deaths from population ageing and growth ([Figure 1A](#)).

Demographic shifts have exerted a progressively evolving impact across three distinct periods from 1990 to 2021 ([Figure 1B](#)). In the initial decade (1990–99), population growth contributed more to rising CVD deaths than the change in age structure from population ageing. In the subsequent decade from 2000 to 2009, population ageing became the primary driver, a pattern that continued through 2010–21. This change aligns with China's transition to an ageing society in 2000, when the proportion of individuals aged ≥ 65 years exceeded 7% (see [Supplementary data online, Figure S3](#)).

As a prediction, the ageing population will continue to grow with a more profound impact on the age structure. Therefore, the number of CVD deaths and crude CVD mortality in China will inevitably increase unless there is a more substantial decrease in the incidence of CVD and case fatality. Several countries have demonstrated the pattern of decreasing crude CVD mortality against population ageing.

The impact of changes in CVD incidence and fatality on age-specific CVD mortality

The decline in age-specific CVD mortality in China can be attributed to either lower CVD incidence, encompassing fatal and nonfatal events across different age groups, or reduced fatality rates, defined as the proportion of fatal CVD cases among all incident events. Lower CVD incidence reflects effective primary prevention and management of major risk factors and recurrent risk management for CVD patients in secondary prevention. The declining proportion of fatal cases largely results from improved capacity and quality of emergency care for acute CVD events, along with advances in secondary prevention for chronic CVD patients.

Previous studies have provided evidence regarding the increasing incidence and decreasing case fatality of CVD and its subtypes in China.^{44–46} However, the contributions of these changes to age-specific CVD mortality and their interrelated effects remain unclear. Comparing changes in age-specific incidence rates and mortality-to-incidence ratio with age-specific mortality rates ([Figure 2](#) and [Supplementary data online, Figure S4](#)), there are nearly synchronized changes in the mortality-to-incidence ratio (reflecting the case fatality rate, Methods in [Supplementary Material](#)) of CVD and its subtypes, mirroring patterns in age-specific mortality. The decline in age-specific IHD mortality among young and middle-aged women, and in ischaemic stroke mortality across both sexes, is

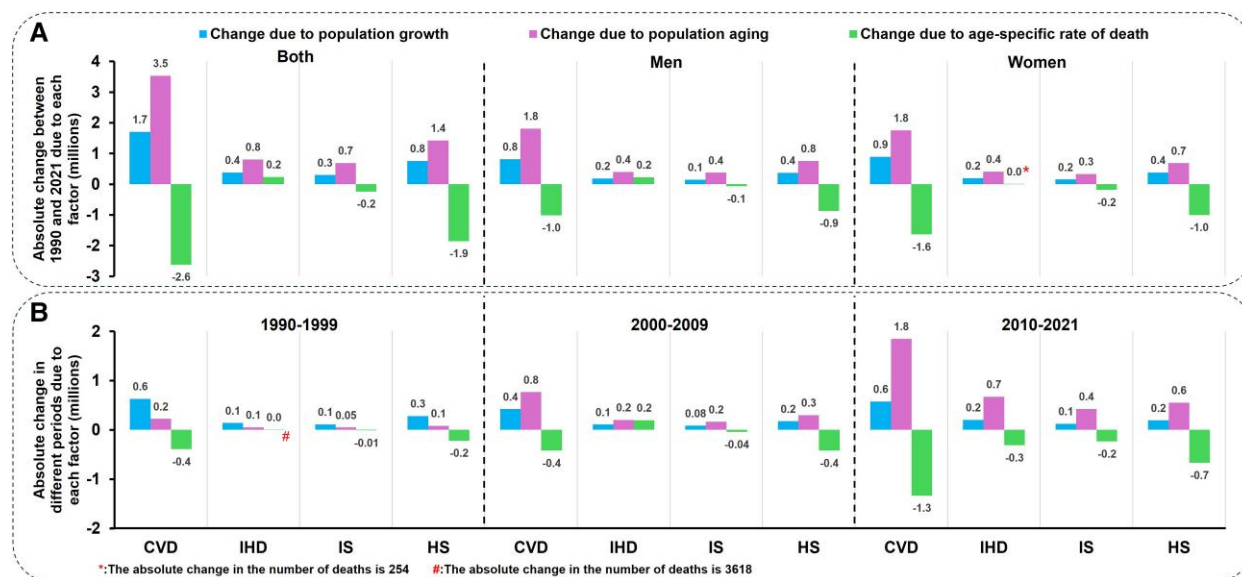


Figure 1 Contribution of changes in population growth, population ageing, and rates of age-specific cardiovascular death to absolute changes in cardiovascular disease death in China between 1990 and 2021 and across different time frames. (A) Decomposition analysis between 1990 and 2021 for the total population aged ≥ 25 years, men, and women, respectively. (B) Decomposition analysis for the total population aged ≥ 25 years for the periods between 1990 and 1999, 2000 and 2009, as well as 2010 and 2021, respectively. The method of decomposition analysis is based on the methodology described by Roth et al,⁴³ detailed in [Supplementary Materials](#). Data are expressed as absolute changes in the number of deaths, measured in millions. CVD includes IHD, IS, HS, and other types. HS includes intracerebral haemorrhage and subarachnoid haemorrhage. Data for age-, sex- and cause-specific rates of CVD death were sourced from the Global Burden of Disease study (<https://ghdx.healthdata.org/gbd-2021>)³³ CVD, cardiovascular disease; IHD, ischaemic heart disease; IS, ischaemic stroke; HS, haemorrhagic stroke.

largely due to the reduction in fatal cases. For IHD and ischaemic stroke, the incidence trends exhibits an inverse relationship with age-specific mortality changes, suggesting that enhanced medical care drove reduction in age-specific rates of these conditions. In contrast, improved risk factor management was the main driver behind the substantial decline in haemorrhagic stroke.

Furthermore, the increase in age-specific incidence of ischaemic stroke and IHD, along with declining fatality risk between 1990 and 2021, implies a continuous rise in the number of nonfatal cases, collectively contributing to a substantial increase in CVD prevalence. The number of CVD prevalent cases nearly tripled since 1990, reaching 133.8 million across all ages in 2021.³³ The trends in the prevalence of CVD, IHD, and stroke were more closely aligned with their corresponding incidence patterns over the past three decades ([Graphical Abstract](#)).

Changes in the profiles of risk factors and the capacity of medical care

Undoubtedly, the changes in incidence and fatality were associated with the epidemic and management profiles of CVD risk factors, as well as the availability and accessibility of effective medical treatment for acute and chronic CVD in China. The rapid economic growth, population ageing, and urbanization over the past three decades have substantially reshaped the lifestyles and environments and resulted in significant changes in CVD risk factors. Recent publications reported the temporal trend and spatial pattern of multiple risk factors,^{13,14,16–18,20,23,24,27} with particularly marked increases in all major metabolic risk factors, including obesity, hypertension, dyslipidaemia, and diabetes.²⁷ Comparing

changes in risk factor exposure with their attributable deaths enables a comprehensive assessment of their interrelationships and supports effective prevention and control policies in China. The increased incidence of IHD and ischaemic stroke from 1990 to 2020s aligns with a surge in prevalence rates of obesity, diabetes, and elevated non-high-density lipoprotein cholesterol across all age groups (see [Supplementary data online, Figure S5](#) and [Supplementary data online, Figure S6](#)). On the other hand, the decreased incidence of haemorrhagic stroke can be attributed to the reduced age-specific prevalence of hypertension over the past decade⁴⁷ and enhanced age-specific treatment and control rates. However, the marked increase in obesity prevalence continues to be a major public health concern, as it not only directly increases the risk of CVD and its major subtypes, including haemorrhagic stroke, but also indirectly elevates CVD risk through the augmentation of other metabolic risk factors. The burden of metabolic risk factors rose sharply in men than in women, consistent with sex differences in the estimated incidence trends ([Figure 3](#)).

Nevertheless, the decline in fatality across most age groups in CVD patients aged < 65 years reflected enhanced healthcare capacity. Over recent decades, China has made remarkable improvements in the accessibility and quality index for medical care.^{48–55} Improving the quality of care for acute coronary syndrome and acute stroke has become an important government agenda.^{56–63} Key efforts included (i) establishing a national monitoring system for quality of care; (ii) setting up the national goals for various subtypes of CVD; (iii) updating performance measure metrics for acute coronary syndrome and stroke; and (iv) developing the networks of chest pain centres, heart failure centres, atrial fibrillation centres, and network of stroke units.^{37,54,62} For example, the Chest Pain Center Program has played a key role in

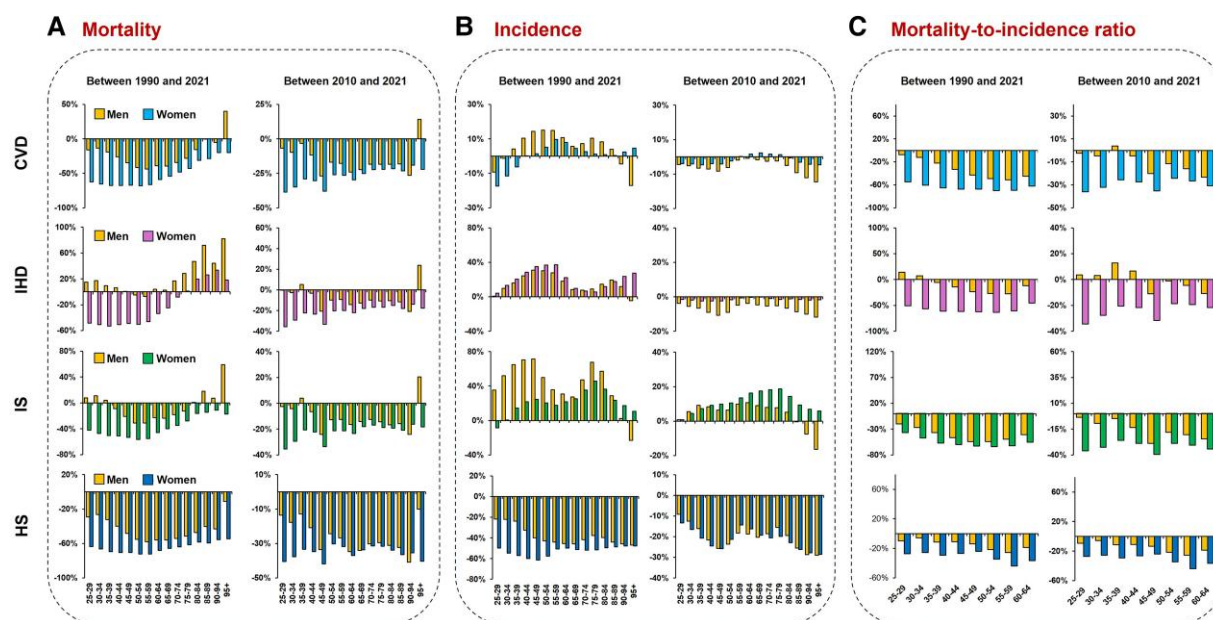


Figure 2 Percentage changes in age-specific mortality (A), incidence (B), and mortality-to-incidence ratio (C) from CVD and its major subtypes between 1990 and 2021 and between 2010 and 2021. HS includes intracerebral haemorrhage and subarachnoid haemorrhage. The ratio of mortality to incidence, as an alternative indicator of case fatality rate, was calculated as the proportion of mortality within a specific age group relative to the corresponding incidence rate of a particular disease over a 1-year period. The percentage change in the age-, sex-, cause-specific mortality, incidence, or MIR rate in 2021 relative to the reference year (1990 or 2010) was calculated as $(\text{rate}_{2021} - \text{rate}_{\text{reference year}}) / \text{rate}_{\text{reference year}} \times 100\%$. Data on MIR are only presented for individuals aged <65 years. Data for age-, sex- and cause-specific mortality and incidence were sourced from the Global Burden of Disease study (<http://ghdx.healthdata.org>).³³ CVD, cardiovascular disease; IHD, ischaemic heart disease; IS, ischaemic stroke; HS, haemorrhagic stroke; MIR, mortality-to-incidence ratio

improving the quality of care for acute coronary syndrome and secondary prevention,⁵⁴ covering 99% of the regions in China by 2024. *Figure 4* exhibits recent progress in quality of care for acute myocardial infarction and stroke.^{57–59,61,63}

Implications of the evolutive features in CVD epidemiology

Challenges from the ageing population and ageing CVD patients

Population ageing is an important underlying cause of increasing CVD death numbers and elevated crude mortality in China.^{40,64} Over the three decades, 3.5 million CVD deaths were attributable to the shifted age structure, completely counteracting reductions from extensive prevention and treatment efforts, and maintaining an increasing trend in crude CVD mortality. This burden is expected to persist as population ageing intensifies in China.⁴⁰ However, the specific impact of ageing on increasing CVD burden is generally considered immutable because ageing is an unmodifiable risk factor. Most CVD epidemiological studies focused on age-standardized rates.^{4,6–8} Emerging evidence suggests that although chronological age is unmodifiable, the intrinsic biological ageing process is modifiable and interacts with other risk factors.^{64,65} The marked decrease in age-specific CVD mortality across all ages, particularly among older individuals in Japan and South Korea—two high-income East Asia countries with more advanced population ageing

but lower crude and age-standardized CVD mortality rates than China—serves as a significant example.^{66–70} Japan's crude CVD mortality in Japan remained stable from 1990 to 1999 and then declined slowly in men during the last 20 years, while women exhibited a pattern similar to that of China throughout the entire period (*Figure 5*). Despite population ageing being approximately 30 years ahead of China, Japan saw continuous declines in age-standardized CVD mortality for both men and women. South Korea experienced a significant decline in age-standardized CVD mortality, with consistently lower and less variable crude CVD mortality rate than Japan or China. The improvements in both South Korea and Japan were primarily attributed to reduced incidence of IHD and stroke through enhanced primary prevention strategies. The substantial decline in haemorrhagic stroke incidence and mortality across all age groups from 25 to ≥95 years over three decades in China further indicates that the impact of ageing is modifiable through improved risk factor control in primary prevention and effective treatment. This suggests China can further mitigate CVD burden by scaling up comprehensive strategies for prevention and treatment nationwide.

Meanwhile, the ageing population has led to an increasing number of older individuals with high CVD risk from multiple risk factors or with acute or chronic CVD. However, evidence on specific strategies in primary prevention, acute care, and secondary prevention for patients over 75 or 80 years of age remain limited,^{71–73} as most evidence for the management of CVD, such as acute coronary syndrome and acute stroke, has been derived from randomized controlled trials (RCTs) that included fewer patients aged over 75 and excluded those with

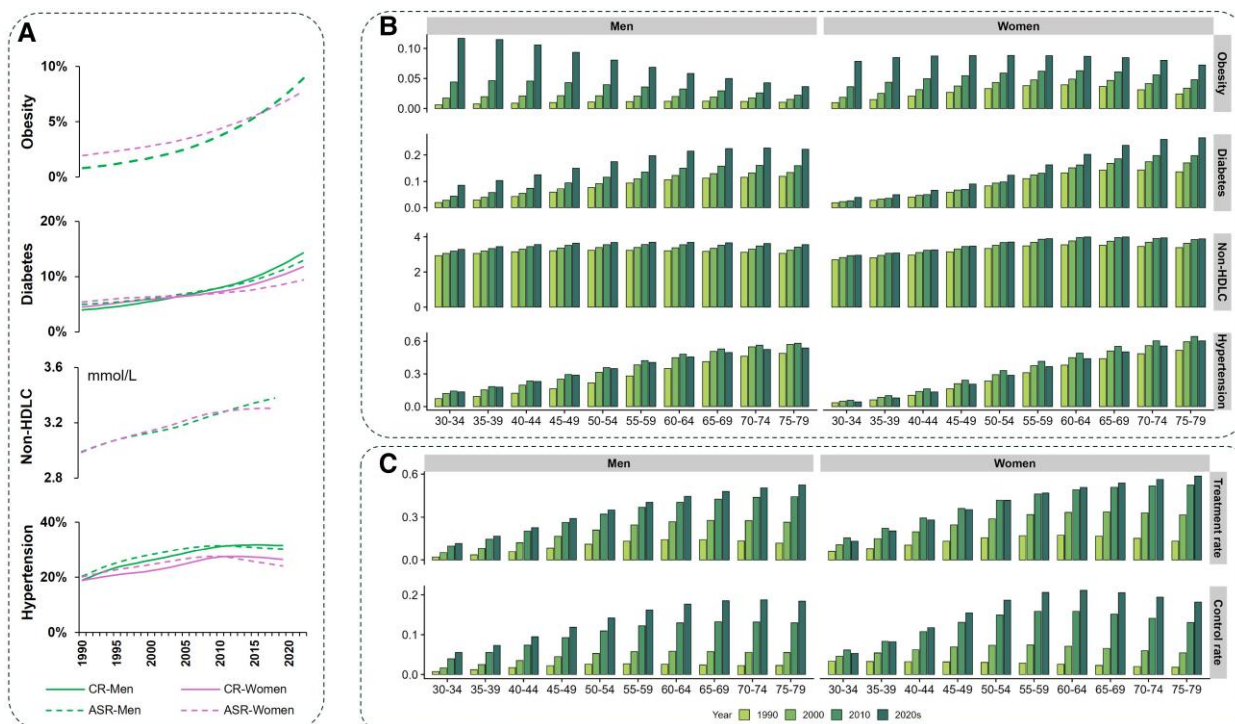


Figure 3 Changes in crude, age-standardized, and age-specific rates or levels of metabolic risk factors from 1990 to 2020s. Data and definitions of metabolic risk factors were derived from the NCD Risk Factor Collaboration (<https://ncdrisc.org/index.html>).³⁴ The latest data on obesity, diabetes, non-HDL-C, and hypertension were sourced from data in 2022, 2022, 2018, and 2019, respectively. Panel A indicates changes in crude and age-standardized rates or levels of metabolic risk factors from 1990 to 2020s. Panel B represents changes in age-specific rates or levels of metabolic risk factors in the year 1990, 2000, 2010, and 2020s. Panel C presents changes in age-specific data on the treatment and control rates of hypertension in the years 1990, 2000, 2010, and 2020s. Non-HDL-C, non-high-density lipoprotein cholesterol; CR, crude rate; ASR, age-standardized rate

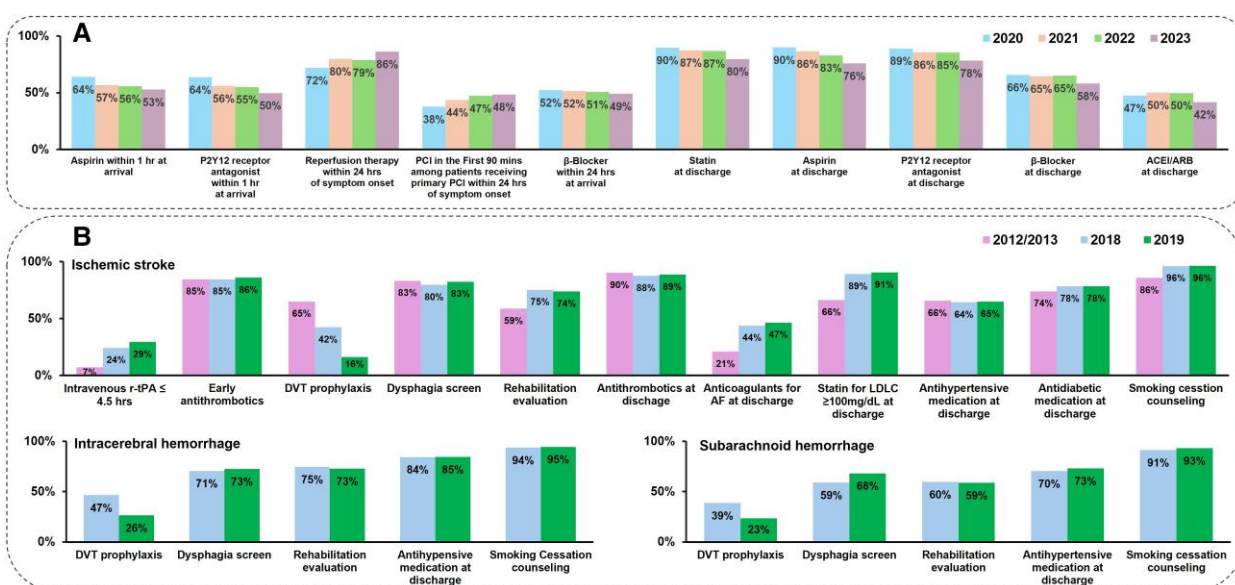


Figure 4 Improvement in ST-segment elevation myocardial infarction and stroke care quality displayed as the percentage of eligible patients who received each guideline-recommended performance metric over the recent decade. Data for ST-segment elevation myocardial infarction indicators (A) were derived from the national report by the National Centre for Cardiovascular Quality Improvement.⁶³ Data for stroke indicators (B) were sourced from the study by Jia *et al.*,⁵⁷ Li *et al.*⁵⁸ and Wang *et al.*^{59,61} PCI, percutaneous coronary intervention; ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; r-tPA, recombinant tissue-type plasminogen activator; DVT, deep vein thrombosis; AF, atrial fibrillation; LDL-C, low-density lipoprotein cholesterol.

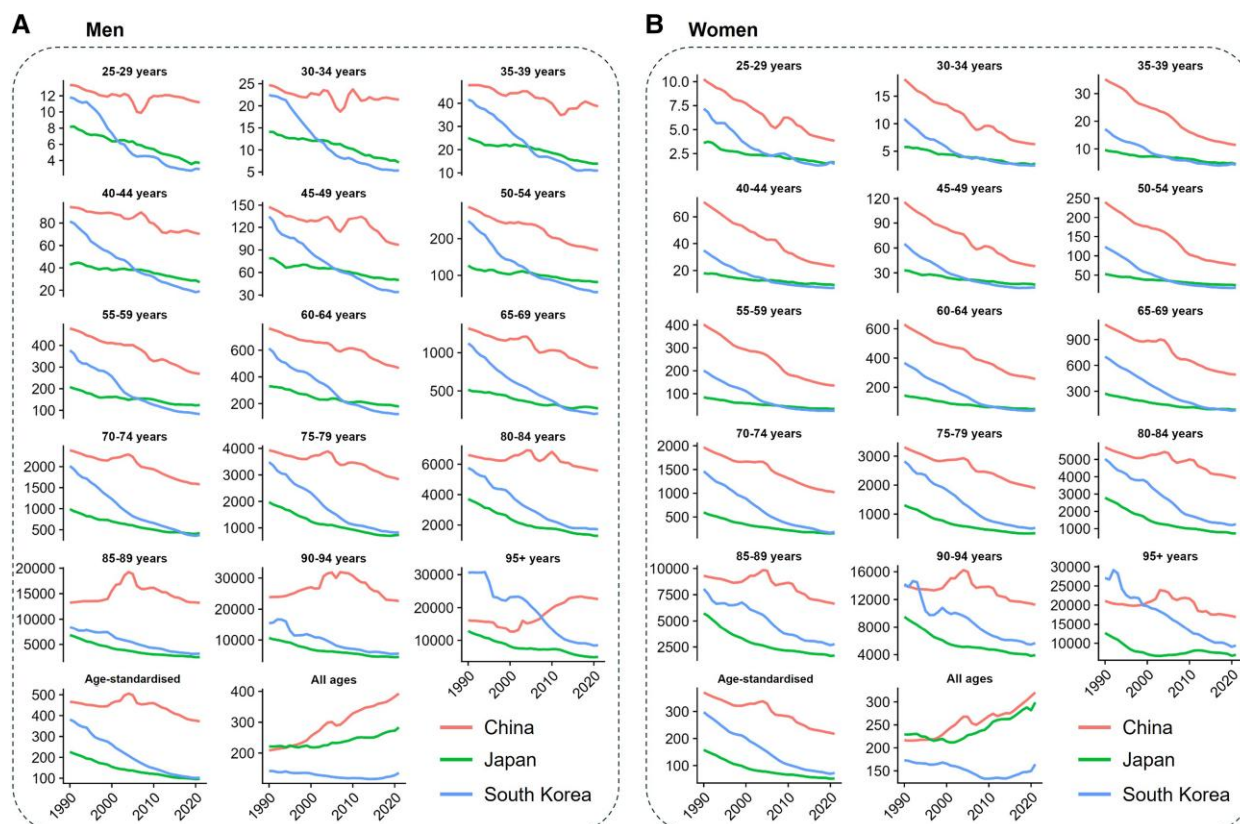


Figure 5 Comparison of crude, age-standardized, and age-specific mortality of CVD from 1990 to 2021 among men (A) and women (B) between China and selected countries in Asia. Data for crude, age-standardized, age- and sex-specific mortality of CVD (per 100 000 population) were sourced from the Global Burden of Disease study (<https://ghdx.healthdata.org/gbd-2021>).³³ CVD, cardiovascular disease

multi-morbidity or frailty.^{6,74} Although some RCT evidence exists to guide treatment in ageing patients, most recommendations for elderly patients have lower-level evidence.⁷²

Furthermore, advances in understanding biological ageing process offer new insights into ageing-related diseases. This includes the identification of ageing-specific biomarkers and functional parameters and translation into human-level assessments for early screening and personalized treatment, exploring the molecular mechanisms of ageing and the intricate relationships among hallmarks of biological ageing, as well as designing effective intervention strategies by combining artificial intelligence (AI), big data, and multi-omics technologies.^{75–82} These advances deepen our understanding of ageing and facilitate more precise interventions to assist individuals in maintaining a higher quality of life in their later stages of life, reducing the burden of ageing-related diseases, and achieving optimal healthy ageing.

The implication of the declining trend in CVD mortality across almost all age groups

Although recent studies reported a decreasing trend in age-standardized CVD mortality in China, fewer studies highlighted how significance of the declining age-specific mortality of CVD and its subtypes.

Furthermore, the comparable declining trends in age-specific mortality of CVD subtypes carry different implications. Only the decrease in haemorrhagic stroke mortality was primarily due to declining incidence

rates in both men and women, effectively reducing the CVD burden. Despite similar mortality reductions observed in IHD and ischaemic stroke, it was largely attributable to decreased fatality rates owing to the enhanced capacity of medical care during this period. Consequently, the reduced mortality from IHD and ischaemic stroke resulted in an increase in nonfatal cases and a growing demand for resources allocated to secondary prevention and rehabilitation.⁸³

The roles of primordial, primary and secondary prevention

The decline in age-specific CVD mortality during the last 30 years was primarily attributable to significant improvements in healthcare infrastructures, medical insurance coverage, and the availability and accessibility of effective treatment strategies for patients with CVD.^{48,49,53,59,61,62} However, unhealthy lifestyles have become deeply ingrained in the modernized patterns of daily living and working environments for the entire population,^{9,12–17,84–88} it has posed severe challenges for primary prevention, resulting in substantially increased prevalence of metabolic risk factors in all ages of both sexes between 1990 and 2021.^{4,17,18,22–24,27,89,90} This trend persists despite a notable decrease in the prevalence of hypertension in women in recent years and some improvement in hypertension treatment and control.^{25,91}

The concurrent rise in the epidemic of metabolic risk factors has led to an increased incidence of atherosclerotic CVD, notwithstanding advances in treatments for acute coronary syndrome and ischaemic

stroke. Multiple national investigations have demonstrated that the widespread adoption of unhealthy lifestyles among children, adolescents, and adults is the primary driver of metabolic risk factor epidemics in China.^{86–90,92–95} A comprehensive national survey revealed that a mere 1.7% of children and adolescents exhibited optimal levels across all seven indicators of ideal cardiovascular health, while only 29% maintained ideal diets and sufficient physical activity.⁹³ Another survey indicated that a mere 0.2% (0.1% in men and 0.3% in women) of Chinese adults achieved ideal cardiovascular health,⁹⁵ further rising incidence of atherosclerotic CVD.

Meanwhile, the decreased case fatality rate in patients with acute CVD events has resulted in an increased prevalence of chronic CVD patients. Notably, more stroke and IHD patients survived their first-ever events,^{96,97} forming a large high-risk group for recurrence^{37,98} and comorbidities.⁹⁹ According to the national report, approximately 13% of patients with ischaemic stroke experienced recurrence within 12 months¹⁰⁰ and 41% within 5 years.⁹⁶ Shared pathophysiological mechanisms and overlapping exposure to multiple risk factors contribute to the incidence and co-occurrence of IHD and ischaemic stroke.⁹⁹ Valid and comprehensive data on incidence remain limited for other CVD subtypes, such as atrial fibrillation and heart failure, which also require special attention.^{32,101} As a serious clinical syndrome and an end-stage CVD manifestation, heart failure becomes increasingly prevalent in China due to population ageing, more survivors and longer survival after acute myocardial infarction and chronic IHD, and increasing prevalence of atrial fibrillation¹⁰¹ but remains persistently high mortality and readmission rates.^{102,103} These circumstances have expanded the demand for secondary prevention.^{37,39} Despite advances in medical care, considerable discrepancies persist between guideline-based recommendations and clinical practice, particularly in low performance rates of reperfusion and intravenous thrombolytic therapies and under-prescribing of guideline-recommended medications at discharge and rehabilitation.^{56–59,61,104} These gaps can be attributed to uncertainty in identifying patients with indications, inadequate capacity of medical care providers, difficulties in balancing therapy benefits and risks, limited patients' knowledge about CVD incidence and severity in asymptomatic conditions, and high out-of-pocket costs.^{57,58,105,106} Meanwhile, substantial variability in the delivery of evidence-based measures existed at geographic (urban and rural; eastern, central, and western regions) and hospital-level (province, prefecture and county; secondary and tertiary) in China, especially in low socioeconomic and rural areas due to limited access to, affordability of, and availability of healthcare resources and lack of high-quality care.^{37,54,56,107–110} Only 36% of acute myocardial infarction patients in rural areas received percutaneous coronary intervention, compared with 51% in urban areas.¹¹¹ Cardiac rehabilitation programmes are concentrated in urban areas, while rural regions lack facilities and trained healthcare professionals.¹⁰⁵ As for ischaemic stroke, rates of intravenous thrombolysis and anticoagulation for atrial fibrillation were much lower in central China than in eastern regions.⁵⁷ In addition, the choice of admission healthcare institution can affect patients' adherence to secondary prevention and treatment.¹¹² As a pivotal component of the healthcare system, China's primary healthcare services require further strengthening.^{41,113} A recent standardized patient-based study reported that China's urban community health centres achieved a 40.0% completion rate for recommended questions and examinations, a 42.3% accuracy rate for correctly diagnosing conditions such as unstable angina, and a 76.5% accuracy rate for prescribing appropriate antihypertensive medications.¹¹⁴ Although these performance metrics surpass those of county hospitals, which represent the highest of the three-tiers of China's rural health system,

primary care institutions remain insufficient to fully meet the current healthcare demands. A severe shortage of skilled healthcare professionals and limited capacity to deliver qualified care further compounds the challenges in providing efficient and effective prevention and treatment within the primary healthcare system.^{41,113} Collectively, these factors have deeply impeded the transition of evidence-based guidelines into clinical practice.

In light of these findings, public health policies and resource allocation must prioritize primordial health interventions for children and young individuals, as well as primary prevention strategies for the broader population.^{115–122} Although the initial effects of health initiatives and policies on air quality and tobacco control recently offer promising opportunities for improving cardiovascular health in China, substantial risks persist, primarily due to poor dietary habits and insufficient physical activity, underscoring the critical need for the implementation of more effective population-level healthy behaviour interventions. A critical component of these efforts should involve enhancing the accessibility and availability of effective, comprehensible, and applicable health information and environments for individuals from diverse educational backgrounds and economic conditions.¹²³ While primordial and primary prevention should be prioritized as strategies for long-term CVD prevention in China, there are also substantial requirements among patients with CVD for further reductions in case fatality rates, recurrence rates, and complications. These reductions can be achieved by enhanced pre-hospital heightened surveillance and emergency care,¹²⁴ the innovation and implementation of effective acute treatment strategies,^{125,126} improved prescriptions of guideline-directed discharge medications,⁵⁶ increased the availability and accessibility of rehabilitation services for secondary prevention,¹²⁷ the extension of high-quality resources and medical care to underserved regions and populations,¹²⁸ the development of localized prevention targets and strategies, as well as the cultivation of skilled practitioners and enhanced alignment of primary care with hospital care.⁴¹

Summary

This review aimed to elucidate the transitional epidemiological characteristics of CVD in China over the past three decades, provide insight into the negative or positive impact of underlying factors that shaped these features (*Graphical Abstract*), and identify implications for emerging challenges and recommendations on strategies and actions in CVD prevention and treatment in China. The perspectives presented in this review may also have applicability to CVD prevention in other countries.

However, the lack of accurate data for the annual incidence and case fatality of major types of CVD, such as atrial fibrillation and heart failure, is an essential limitation in epidemiology studies worldwide. With the rapid development of AI technology, there is an increasing possibility of establishing an AI-assisted surveillance system that can interconnect all the health-related data sources and provide timely information on CVD epidemiology in the future.

Supplementary data

Supplementary data are available at the *European Heart Journal* online.

Declarations

Disclosure of Interest

All authors declare no disclosure of interest for this contribution.

Data Availability

The data utilized in this study are available from the Global Burden of Disease study (<http://ghdx.healthdata.org>), NCD Risk Factor Collaboration (<https://ncdrisc.org/index.html>), and other published documents.

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Nothing to declare.

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